Risk

Identifying, analysis and managing risk

Risk

- Risks in software development
- Risk management
- Risk management plan
- Risk monitoring and mitigation action
- Agile risk management

Risk in Software Development

Project risk

- Software development involves technological advancements
- Strive for new features, efficiencies or exploiting advancements
- Requires high levels of knowledge
- As such software development project contains uncertainty
- This is known as project risk

Risk exposure

- Risk exposure is the probability and size of the potential failure
- High levels of risk exposure can compromise project success
- Risk exposure represents a possibility of project failure

Risk analysis and management

- Identifying risk
- Classifying risk
- Quantifying the risk exposure
- Removing or reducing risk exposure

Types of risk

- Most software projects can have four risk impact areas
  - Unproven technologies
  - User and functional requirements
  - Application and system architecture
  - Performance

Unproven technology risks

- New tools, techniques, protocols, standards, & development systems
- These increase the probability that technology risks will arise
- Improper use of new technology can lead to project failure
- Training and knowledge are critical in addressing this type of risk
- Spike stories can help to address this type of risk

User and functional requirements risks

- User needs lead to user and functional requirements
- Needs can change during a project due to knowledge discovery
- These changes can propagate throughout a project’s requirements
- These disruptions can lead to failures in a poorly planned project
- Agile is good at avoiding and addressing this type of risk

Application and system architecture risks

- Taking the wrong direction with system design can lead to failures
- Understanding application or system architecture is vital to success
- Lack of an overarching architecture is a real risk in poorly managed agile projects
- Knowledge will inform making sound design choices
- Spike stories can help to address this area of risk

Performance risks

- Risk management must encompasses expectations of performance
- Need to ensure products are moving in the right direction
- Establishing business rules or cinsteiants, or NFRs can help

Risk Management

Risk Management goals

- Risk management means risk containment and mitigation, seldom removal
- First identify risks and plan against them
- Then be ready to act when a risk arises
- This will help to minimise the impact to the project

Risk Management steps

- Identify risks and their triggers
- Classify the risks
- Quantify the risk exposure
- Prioritise the risks and the associated risk response
- Create a plan that links each risk to a mitigating action
- Monitor for risk triggers during the project
- Implement the mitigating action if any risk materialises
- Communicate risk status throughout project
- Some methodologies use this information to help priorities requirements

Risk Management Plan

Risk Management Plan

- Create a risk management plan
- Catalogue risks according to type and level
- Specify the response for each risk
- The following is a very simple example of a risk catalogue
  - Often use more columns to model probability and impact

RISK	LEVEL	ELIMINATION	BACKUP
Get Invited API	10/High	This risk is the first to be tackled, as it will be used for the applications main purpose of pulling in events. A prototype will be developed using the Get Invited API, thus removing any concerns about developing with it. Without this functionality the application will not be functional.	If the risk can't be eliminated, and the API becomes too complex to work with there are many alternatives. EventBrite is the most common with many online tutorials so it will be used as a suitable replacement.
Secure login	8/High	Having a secure login for the application not only provides the platform in which the app will operate but also provides user trust for the application. Apps without secure logins do not have a user base. The developer has developed many secured login systems therefore having a reference point as well as sufficient knowledge.	A basic login system can be used as a back up until a more secure login can be developed. The application can be made as a minimum viable product and therefore not require an extremely secure interface.

		project. Creating user connections has been done previously. With the chosen language of PHP not only will there be a base knowledge but a large community online of people making similar functionally.	functionality will be achieved. Allowing for extra time will insure that it is completed.
Inviting friends to events.	5/ Intermediate	Once the previous risk of being connected to friends within the application is completed the next most important feature is the ability to invite these friends to the chosen event. Tutorials online of how many people have implemented this functionality exist.	This risk needs to be developed and an alternative can't be suggested. Allowing suitable time to develop this will help eliminate the risk.
Profile Creation	4/Low	Profile creation will allow the user to personalise their experience within the application. Queries to a database, to update user information has been completed on previous projects.	If the profile creation falls out of scope with time spent on higher risks the alternative will be to provide a more basic profile system with less customisation for the user.

Risk Management & Mitigation

Risk Monitoring and Mitigation action

- Risk monitoring has to be integral to the project
- This means frequent checking during project meetings and critical events
- When a risk occurs the planned mitigation response should be taken

Risk Mitigation options

- Accept
  - Acknowledge that a risk is impacting the project
  - Make an explicit decision to accept the risk without any changes to the project
- Avoid
  - Adjust project scope, schedule, or constraints to minimise the effects of the risk
- Control
  - Take action to minimise the impact or to reduce intensification of the risk
- Monitoring
  - Monitor the project environment for potentially increasing impact of the risk
  - Only suitable for low-impact risks

Prototyping to manage risk

What to prototype

- Targets a significant risk related to the product for exploration
  - Likely the biggest technical obstacle to project success
- Perhaps related to
  - Key user and functional requirements
  - Core feature or data structure
  - New or unproven technologies
  - Technology that is new to the team
- Prototyping risk out is a common, traditional, response
  - Might remove the targeted risk
  - At worst, helps to understand the risk better

Prototypes

- A prototype is not finished software nor an MVP
  - Focused on addressing a risk
  - Low fidelity and most likely throw-away
  - Very rarely is the appearance of significance
  - Almost never focused on customer value
- Prototype from an informed position, based on analysis and research
- Review to determine what has been learned
- Allow for a few false starts and dead ends

Traditional approach pros and cons

Pros and cons of traditional risk management

- Pros
  - Risks identified early before major time or money investment
  - Might reduce chances of only discovering risks in the middle of the project work
  - Advance contingency planning helps the team move faster in the face of a risk
  - Whole team has sight of the identified risks at the outset
- Cons
  - Not traditionally reviewed or updated during the project
  - Often not linked directly to requirements, although can be
  - Not always visible to the whole team, depending on methodology

Risks management in Agile

Agile inherently protects against risk

- Underlying philosophy accepts that not everything is known and allows for change
- Structured to react to the unknown through learning and adaptation
- Continuously reviews what is needed and how to deliver it
- Can continuously look for risks during the project, not just as an upfront task
- Transparency, helps to make risks visible early, and keep them visible

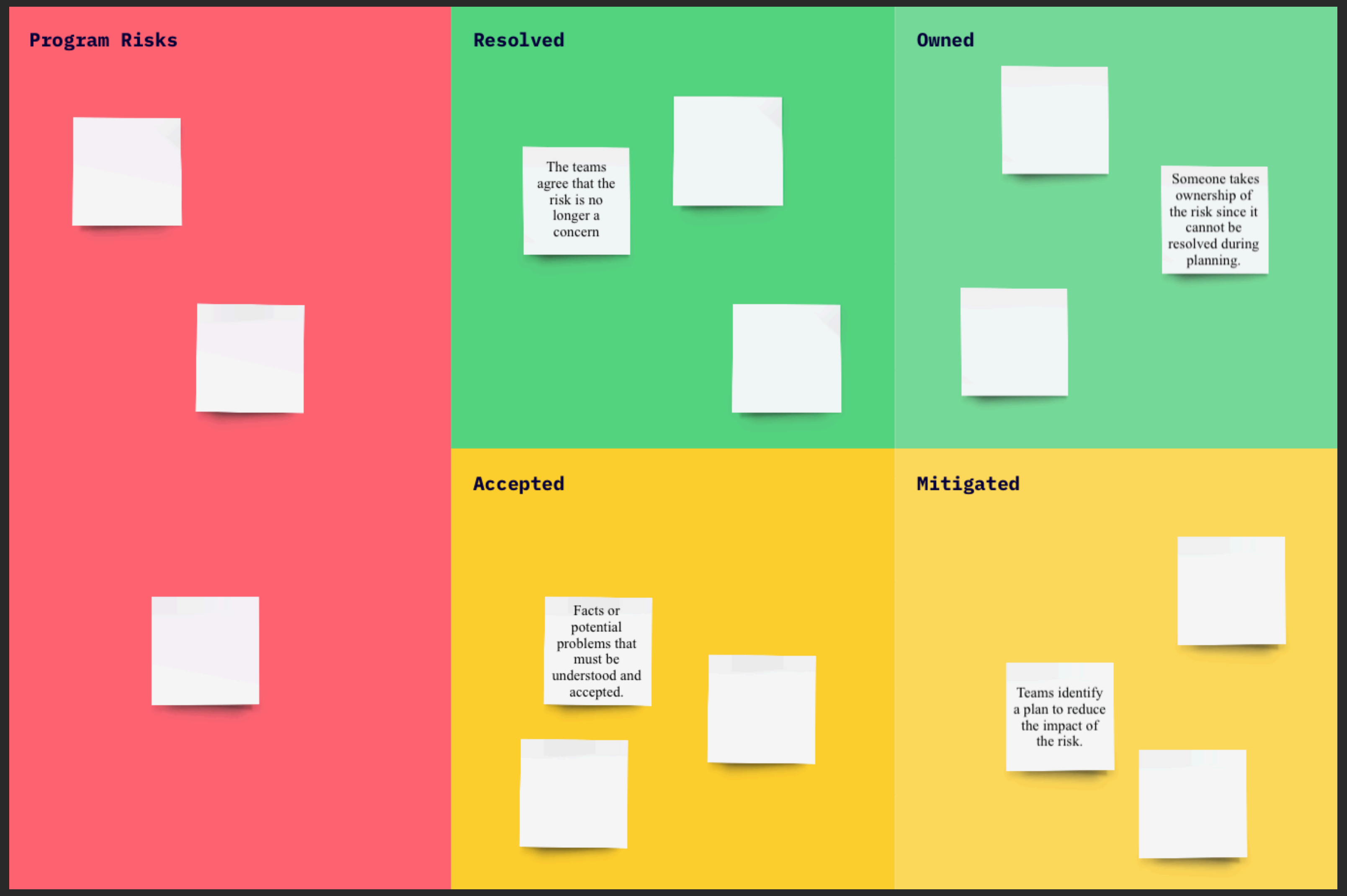
Core Agile practices to protect against risk

- Collaborative planning, wider expertise makes identifying risks more likely
- Customer consultation, helps ensure the correct product is delivered
- Daily standups, help to expose and consider risks as they emerge
- Story and backlog reviews, helps apply emerging understanding to likely areas of risk
- Story estimation, helps to explore the nature of stories, including any risks
- Story splitting, helps to remove areas within which risk can be hidden
- Spike stories, can explore specific risks in order to better understand them
- MVPs, have the benefits of prototyping but also brings value and data at each release

Additional Agile practices to protect against risk

- Risk burn down chart
  - Simple visual indicator of risk trends within the project
  - Shows risk exposure, impact and response
  - Shows progress towards controlling risks

- Risk board
  - Makes risks transparent to team and other stakeholders
  - Displays risks, occurrence probabilities and likely impacts
  - Reviewed during stand-up meetings, on a daily basis
  - A ROAM Board, from SAFe is an example of this



Program Risks

Resolved

The teams agree that the risk is no longer a concern

Owned

Someone takes ownership of the risk since it cannot be resolved during planning.

Accepted

Facts or potential problems that must be understood and accepted.

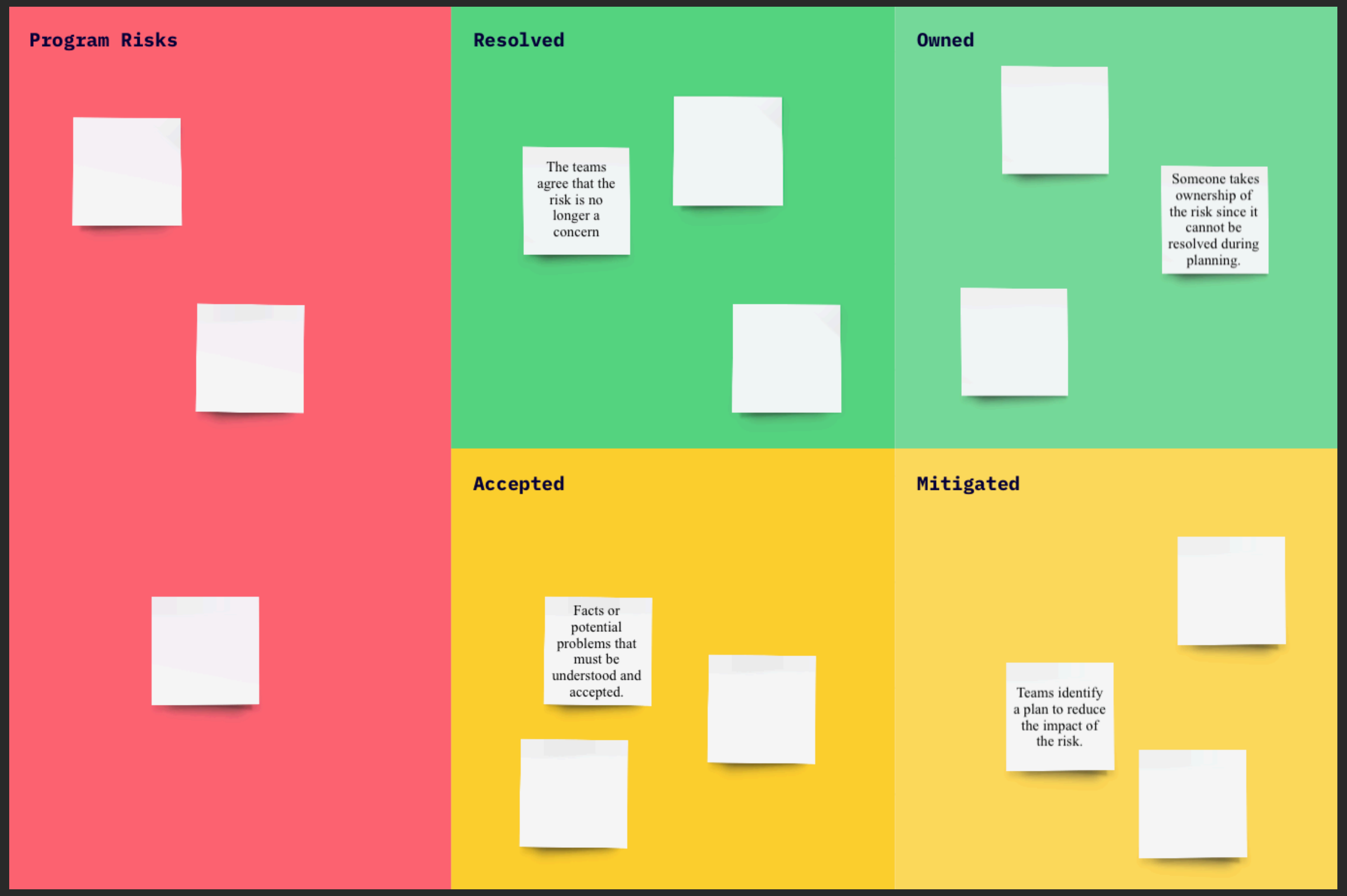
Mitigated

Teams identify a plan to reduce the impact of the risk.

ROAM Board

- Program Risks: Unclassified risks identified during Program Increment (PI) planning
  - PI is a SAFe term, similar to Product Backlog Refinement in Agile Scrum
- Resolved: risk no longer an issue
- Owned: a team member has taken ownerships of the risk
- Accepted: risk cannot be resolved but is understood
- Mitigated: mitigation plan is in place to manage the risk

- Process
  - Identify risks during PI planning and add to Program Risks
  - Categorise each risk
  - Vote as a team to prioritise each risk
  - Update ROAM board during future review meetings



Blended approach

- Traditional risk management and Agile techniques can be complementary
- Many teams make use of traditional and agile specific approaches for risk management
  - Or select which to emphasis depending on the nature of the project or the client
- Some projects might require formal methods due to legal requirements
  - To fit with compliance models
  - To service domain specific documentation requirements

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